

SECSR2043 OPERATING SYSTEMS

[20 Marks]

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 Section : 02

Marks

Instruction: Please answer all of the following questions. Whenever the 🙋 symbol appears, please raise your hand to call your instructor, he/she will verify your results by putting his / her initial next to the symbol.

1. Type the following commands using a text editor and save it as a *yourname.sh* (Example: ahmad.sh).

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt >
orange.txt
cd drinks; cp ../cars/*.sh .; cp ../fruits/*.sh .;
cp ../*.jar .
```

- a) Execute the script and draw a tree structure that contains created directories and files. The parent node of the directory begin with **\$HOME** directory.

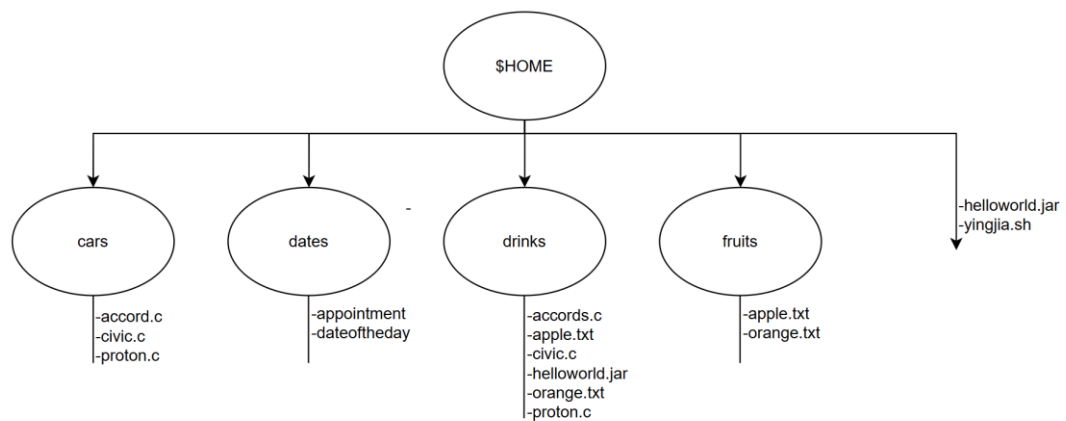
[4 marks]



Print screen the script that you type;

```
echo "Hello world" > helloworld.jar
mkdir cars; mkdir dates; mkdir fruits drinks
cd cars; echo "Honda Accord" > accord.c
cp accord.c civic.c; echo proton > proton.c; cd ../dates;
date > dateoftheday
cat dateoftheday > appointment
cd ../fruits; echo apple > apple.txt; cat apple.txt > orange.txt
cd ../drinks; cp ../cars/*.c .; cp ../fruits/*.c .;
cp ../*.jar .
```

Then draw the tree



```
chau@secre2043:~/lab3$ tree
.
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
|-- helloworld.jar
`-- yingjia.sh

5 directories, 15 files
```



- b) Write an interactive bash script that will read a type of file extension, display all those files, and count the number of files. To validate your script, display `c` program files, and enter “`c`” as the input to the bash script. [4 marks]

Print screen the bash script you type and run

```
GNU nano 7.2 ifiles.sh
#!/bin/bash

# Prompt user to enter file extension
read -p "Enter file extension (without dot): " ext

# Display matching files
echo "Files with ${ext} extension:"
find . -type f -name "*.${ext}"

# Count how many files with the given extension were found
count=$(find . -type f -name "*.${ext}" | wc -l)
echo "Total number of files with ${ext} extension: $count file(s) found"
```

```
chau@secr2043:~/lab3$ nano ifiles.sh
chau@secr2043:~/lab3$ chmod +x ifiles.sh
chau@secr2043:~/lab3$ ./ifiles.sh
Enter file extension (without dot): c
Files with .c extension:
./drinks/civic.c
./drinks/accord.c
./drinks/proton.c
./cars/civic.c
./cars/accord.c
./cars/proton.c
Total number of files with .c extension: 6 file(s) found
```



2. The following Figure 1 illustrates a tree structure of some directories and files.

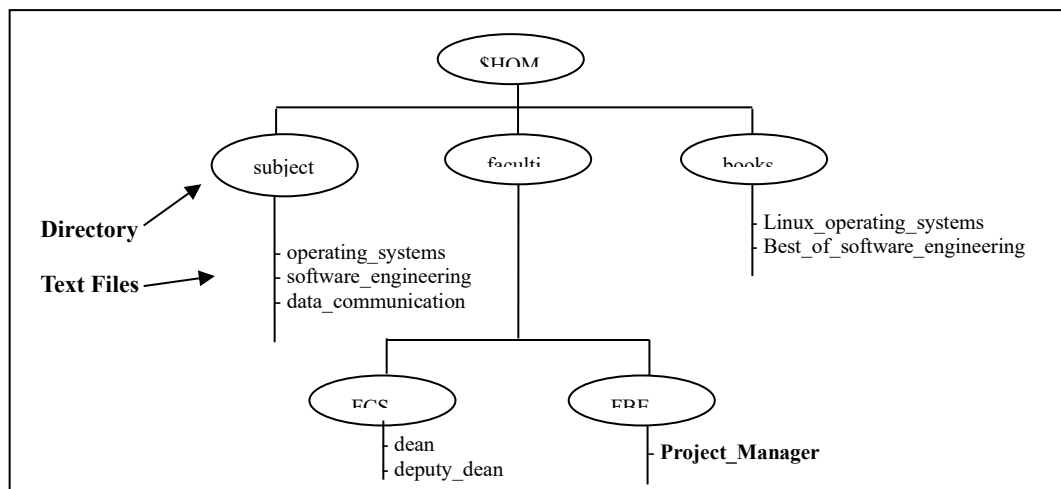


Figure 1

- a) Write a bash script (called `myname2a.sh`) that will produce directories and files as in Figure 1. Each text files contain its filename without the underscore character. For example: text file `Project_Manager` contains `Project Manager`). [4 marks]

Print screen the bash script you type and run

```
GNU nano 7.2 yingjia2a.sh
#!/bin/bash

mkdir subject; mkdir faculties; mkdir books

echo "operating systems" > subject/operating_systems
echo "software engineering" > subject/software-engineering
echo "data communication" > subject/data_communication

mkdir faculties/FCS; mkdir faculties/FBE

echo "dean" > faculties/FCS/dean
echo "deputy dean" > faculties/FCS/deputy_dean

echo "Project Manager" > faculties/FBE/Project_Manager

echo "Linux operating systems" > books/Linux_operating_systems
echo "Best of software engineering" > books/Best_of_software_engineering
```

```
chau@secr2043:~/lab3$ nano yingjia2a.sh
chau@secr2043:~/lab3$ chmod u+x yingjia2a.sh
chau@secr2043:~/lab3$ ./yingjia2a.sh
chau@secr2043:~/lab3$ tree
.
|-- books
|   |-- Best_of_software_engineering
|   `-- Linux_operating_systems
|-- cars
|   |-- accord.c
|   |-- civic.c
|   `-- proton.c
|-- dates
|   |-- appointment
|   `-- dateoftheday
|-- drinks
|   |-- accord.c
|   |-- apple.txt
|   |-- civic.c
|   |-- helloworld.jar
|   |-- orange.txt
|   `-- proton.c
|-- faculties
|   |-- FBE
|   |   |-- Project_Manager
|   `-- FCS
|       |-- dean
|       `-- deputy_dean
|-- fruits
|   |-- apple.txt
|   `-- orange.txt
|-- helloworld.jar
|-- ifiles.sh
|-- subject
|   |-- data_communication
|   |-- operating_systems
|   `-- software-engineering
|-- yingjia.sh
|-- yingjia2a.sh
10 directories, 25 files
```



- b) Complete the following table by writing the access control of directories or files that were produced. Given is the access control for directory called book.

[2 marks]

Directory/File	Access Control
books	drwxrwxr-x
subjects	drwxr-xr-x
Best_of_software_engineering	-rw-rw-r--
FCS	drwxrwxr-x
project_manager	-rw-rw-r--



```
drwxrwxr-x 2 chau chau 4096 Jun 15 12:05 books
drwxr-xr-x 2 chau chau 4096 Jun 15 12:05 subject
-rw-rw-r-- 1 chau chau 29 Jun 15 12:05 Best_of_software_engineering
drwxrwxr-x 2 chau chau 4096 Jun 15 12:05 faculties/FCS
-rw-rw-r-- 1 chau chau 16 Jun 15 12:05 faculties/FBE/Project_Manager
```

- c) Write another bash script (called `myname2c.sh`) that will change the access control of the directories and files based on the following information:

[4 marks]

Directory/File	Users								
	Owner			Group			Public		
subjects	✓	✓	✓	✓	x	x	✓	x	x
Best_of_software_engineering	✓	x	✓	x	✓	x	x	x	x
FCS	✓	✓	x	x	x	x	✓	✓	✓
project_manager	x	x	x	x	✓	✓	x	x	✓

Print screen the bash script you type and run

```
GNU nano 7.2 yingjia2c.sh
#!/bin/bash

# Set permission for subject
chmod 744 subject

# Set permission for Best_of_Software_engineering
chmod 520 books/Best_of_software_engineering

# Set permission for FCS
chmod 607 faculties/FCS

# Set permission for project_manager
chmod 031 faculties/FBE/Project_Manager
```

```
chau@secr2043:~/lab3$ nano yingjia2c.sh
chau@secr2043:~/lab3$ chmod u+x yingjia2c.sh
chau@secr2043:~/lab3$ ./yingjia2c.sh
chau@secr2043:~/lab3$ ls -ld subject
drwxr--r-- 2 chau chau 4096 Jun 15 12:05 subject
chau@secr2043:~/lab3$ ls -l books/Best_of_software_engineering
-r-x-w---- 1 chau chau 29 Jun 15 12:05 books/Best_of_software_engineering
chau@secr2043:~/lab3$ ls -ld faculties/FCS
drw----rwx 2 chau chau 4096 Jun 15 12:05 faculties/FCS
chau@secr2043:~/lab3$ ls -l faculties/FBE/Project_Manager
-----wx--x 1 chau chau 16 Jun 15 12:05 faculties/FBE/Project_Manager
```



- d) Complete the following table by writing the access control for each directory or file after executing the bash script in question 2(c)). [2 marks]

Directory/File	Access Control
subjects	drwxr--r--
Best_of_software_engineering	-r-x-w----
FCS	drw----rwx
project_manager	-----wx--x

End of Lab 3

**** All the Best for Final Exam ****